

Topic: Cartesian Product Operation

Q. What is a Cartesian Product in DBMS? Explain its working mechanism and provide an example demonstrating how it combines two relations.

📌 Definition to Remember

The **Cartesian Product** (also called Cross Product, denoted $\times$) is a **binary operation** in relational algebra that pairs **every row of the first relation with every row of the second relation**, producing all possible combinations.

Output columns = sum of both columns; Output rows = product of both row counts. Union Compatibility is **NOT** required.

Cartesian Product Operation ($\times$)

- **Binary operation** — operates on two tables (which need NOT be union compatible).
- **Syntax:** Relation_A $\times$ Relation_B
- **Output Degree (Columns):** Columns_A + Columns_B
- **Output Cardinality (Rows):** Rows_A $\times$ Rows_B

2. Working of Cartesian Product

- Takes **every single row from A** and pairs it with **every single row from B**.
- Generates all possible row combinations — most of which are **meaningless on their own**.
- On its own, the Cartesian Product produces massive redundant data.
- It is primarily used as the **first step for JOIN operations**: Apply Cartesian Product $\rightarrow$ Apply Selection (σ) to filter meaningful rows.

```
A  $\times$  B :  
Row1_A  $\leftrightarrow$  Row1_B  
Row1_A  $\leftrightarrow$  Row2_B  
Row2_A  $\leftrightarrow$  Row1_B  
Row2_A  $\leftrightarrow$  Row2_B  
(2  $\times$  2 = 4 combinations)
```

3. Example

COLORS (2 rows)

Color_ID	Color_Name
1	Red
2	Blue

SHAPES (2 rows)

Shape_ID	Shape_Name
A	Circle
B	Square

Query: COLORS × SHAPES

Output: 2 + 2 = **4 columns**; 2 × 2 = **4 rows**

Color_ID	Color_Name	Shape_ID	Shape_Name
1	Red	A	Circle
1	Red	B	Square
2	Blue	A	Circle
2	Blue	B	Square

★ Must-Write Points (for 10 marks)

1. Cartesian Product (×) = Cross Product — pairs every row of A with every row of B.
2. Output columns = Columns_A + Columns_B; Output rows = Rows_A × Rows_B.
3. Union Compatibility is **NOT** required — the two relations can be completely different.
4. On its own, produces large amounts of **meaningless data**.
5. It is the foundation of **JOIN operations**: Cartesian Product + Selection = Join.
6. If A has m rows and B has n rows, the result has m × n rows.
7. Cartesian Product is a basic (fundamental) operator in relational algebra.

⚡ Quick Recall

× (Cartesian Product) → Binary → All Row Combinations → Cols = A+B, Rows = A×B → No Union Compatibility Needed → Foundation of JOIN